

THE KENYA NATIONAL EXAMINATION COUNCIL

KCPE 2004

MATHEMATICS

Time:2 hours

READ THE INSTRUCTION CAREFULLY

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet

HOW TO USE THE ANSWER SHEET

4. Use only ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full index number (i.e. school code number and three figured candidate's number) grid near the top of the answer sheet.
7. Do not mark outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written

Example

In the question booklet :

10. What is the next number in the pattern:

1.2, 2.4, 4.8, _____?

- A. 6.0
- B. 7.2
- C. 8.4
- D. 9.6

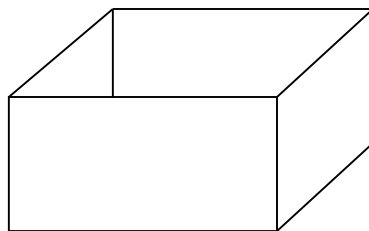
On the answer sheet:The correct answer is D

1. What is 399.90485 rounded off to the nearest hundredth?
 - A. 400
 - B. 399.905
 - C. 399.90
 - D. 399.9
2. What is Greatest Common Divisor (G.C.D) of the numbers 32,48 and 56
 - A. 8
 - B. 168
 - C. 672
 - D. 86016
3. What is three million six thousand four hundred and seventy two less one thousand written in symbols?
 - A. 3599472
 - B. 3005472
 - C. 3007472
 - D. 3059472
4. Which is the correct order of writing the fraction $\frac{3}{8}$, $\frac{5}{16}$, $\frac{11}{32}$, $\frac{13}{48}$ from the largest to the smallest?
 - A. $\frac{13}{48}$, $\frac{11}{32}$, $\frac{5}{16}$, $\frac{3}{8}$
 - B. $\frac{13}{48}$, $\frac{11}{32}$, $\frac{5}{18}$, $\frac{3}{8}$
 - C. $\frac{3}{8}$, $\frac{5}{16}$, $\frac{11}{32}$, $\frac{3}{8}$
 - D. $\frac{3}{8}$, $\frac{11}{32}$, $\frac{5}{16}$, $\frac{13}{48}$
5. What is the place value of the digit 7 in the number 347026?
 - A. Thousands
 - B. Ten thousands
 - C. Seventy thousands
 - D. Hundred thousands
5. In the figure below ST and SR are equal. Angle RPQ = 45° angle PQR = 76°

What is the size of angle PST?

- A. 135°
 - B. 121°
 - C. 118°
 - D. 62°
6. A company sold 2.5 tonnes of rice in 2 kg packets. How many packets were sold?
 - A. 125

- B. 1250
C. 2500
D. 5000
7. Below is the number of tree seedlings planted by each pupil tree planting day:
8 15 11 9 17 14 12 14 13
10 18 8 9 16 13 13 10 12
What was the mean number of seedlings planted by each pupil?
A. $12 \frac{1}{3}$
B. 222
C. $12 \frac{1}{2}$
D. 13
8. What is the value of
 $0.5 + 0.2 \times 4.1 - 0.31$?
0.02
A. 128
B. 60
C. 132.65
D. 50.5
9. A length of 4.6cm on a scale drawing represents an actual length of 9 200m. What is the scale used?
A. 1:200
B. 1:2000
C. 1:20000
D. 1:2000000
10. The diagram below represents an open rectangular trough



- What is the difference between the number of faces and the number of edges?
A. 7
B. 6
C. 5
D. 12
11. What is the value of $\frac{5}{8} + \frac{1}{8} \div \frac{2}{3} - \frac{3}{8}$
A. $1 \frac{3}{56}$

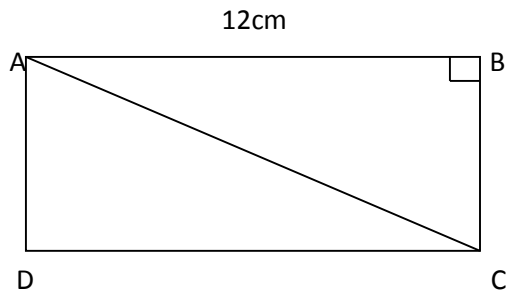
- B. $\frac{3}{4}$
C. $\frac{7}{16}$
D. $\frac{1}{3}$
12. A bag of sugar weighed 108.5kg. Mwangi bought 16 bags. He then removed 0.3kg of sugar from each bag. What total weight of sugar remained?
A. 1736kg
B. 108.2kg
C. 1731.2kg
D. 1735.7kg
13. What is $1\frac{1}{2}\%$ expressed as a ratio in its simplest form?
A. 3:2
B. 2:300
C. 3:200
D. 200:3
14. One hundred and twenty litres of juice is packed in two deciliters packets. How many packets are needed?
A. 60
B. 600
C. 6000
D. 60000
15. Musa bought the following items from a kiosks:
3kg of tomato
 $\frac{1}{2}$ kg of onion @sh. 30
2kg packets of maize flour for sh. 38
 $\frac{1}{2}$ kg of sugar @ sh. 44
He paid for the items using a sh. 200 note. How much balance did he get?
A. Sh. 50
B. Sh. 150
C. Sh.63
D. Sh. 12
16. What is the value of $24 \div 3 + 4 \times 5 - 8 \div 4 \times 10 + 1$?
A. 7
B. 9
C. 41
D. 131
17. The table below shows the cost in shillings of sending parcels through the post.

Weight steps	Africa	Europe Near East	Rest of the world
Up to 1 kg	940.00	1030.00	1070.00
Over 1kg up to 3kg	1335.00	1390.00	1455.00
Over 3kg up to 5kg	165.00	1740.00	1920.00
Over 5 kg up to 10kg	2370.00	2340.00	275.00

Asif sent one parcel weighing 2 kg to Africa and other weighing 3 kg 600g Near East. How much did he spend?

- A. Sh. 3075
- B. Sh.1335
- C. Sh.3030
- D. Sh.2380

18. The figure below shows a rectangular lawn ABCD in which AB = 12m and AC = 15



What is the area of the lawn?

- A. 180m^2
- B. 108m^2
- C. 54m^2
- D. 36m^2

19. What is the next number in the pattern

3, 4, 7, 11, 8 _____?

- A. 19
- B. 25
- C. 43
- D. 29

20. The marked price of an article was sh. 300. Ali paid sh. 260 for the article after being given a discount. What percentage discount did he get?

- A. $86\frac{2}{3}\%$
- B. 40%
- C. $15\frac{5}{13}\%$
- D. $13\frac{1}{3}\%$

21. The figure below represents a vegetable garden bounded by two semi-circle, 5 m apart. The diameter of the larger semi-circle is 30m.

What is the perimeter of the garden?

- A. 78.50m
- B. 88.50m
- C. 83.50m
- D. 96.35M

22. Asha was given a loan of sh. 48 000. She repaid the loan after two years with compound interest at rate of 25% p.a. How much money did she pay altogether?

- A. Sh. 60 000
- B. Sh. 63 000
- C. Sh. 72 000
- D. Sh. 75 000

23. What is the square root of $1\frac{7}{9}$?

- A. $\frac{4}{9}$
- B. $\frac{8}{9}$
- C. $1\frac{1}{3}$
- D. $3\frac{13}{81}$

24. The figure below is made up of a parallelogram NOPP, triangle PRQ and MNR. Angle NMR = 70° and angle NRM = 42°

What is the size of angle PRQ?

- A. 112°
- B. 68°
- C. 44°
- D. 70°

25. A saleswoman was paid a monthly salary of sh. 9000. She was also paid a commission of 10% for all the goods she sold above sh. 10 000. In one month she sold goods worth sh. 25 000. What did she earn at the end of that month?

- A. Sh. 11 500
- B. Sh. 12 500
- C. Sh. 10 500
- D. Sh. 1 500

26. In triangle EFG line EF, angle EFG = $4x^\circ$ and angle EGH = $5x^\circ$

What is the value of angle FEG?

- A. 20
- B. 80°
- C. 100°
- D. 120°

27. Kigen bought goods worth sh. 2400. He gave the shopkeeper three sh. 1 000 notes. He received the balance in equal numbers of sh. 200 and 100 notes.

What total numbers of notes did he receive?

- A. 6
- B. 5
- C. 4
- D. 3

28. The table below shows the number of tonnes of sugar produced and sold by a factory in 6 days

On which was the number of tonnes of sugar sold one and half times the number of tonnes produced?

- A. Thur
- B. Mon
- C. Wed
- D. Sat

29. The number of patients who visited a health centre on Monday was 125, on Tuesday was 163 and on Wednesday was 210.

On Thursday the number was 15 less than those who visited on Wednesday. Equal number of patients visited the centre on Friday and Saturday. The total number of patients visited the centre in six days

How many patients visited the centre on Friday?

- A. 396
- B. 288
- C. 183
- D. 198

30. The area of the curved surface of cylinder is 6160 cm^2 The length of cylinder is 10cm.

What is the radius of the cylinder in centimeters?

(Take $\pi \frac{22}{7}$)

- A. 98
- B. 14
- C. 196
- D. 968

31. The table below shows Mwasi's income from the sale of farm produce, during one year. The information on the income for potatoes is not given.

A pie chart was drawn to represent the information above. If the angle representing the income for beans was 63° What was the income for the potatoes?

- A. Sh. 120 000
- B. Sh. 18 000
- C. Sh. 17 850
- D. Sh. 102 000

32. A bus left Migori town for Nairobi at 19 00h. It arrived in Nairobi 9 hours later. At what time did the bus arrive in Nairobi?

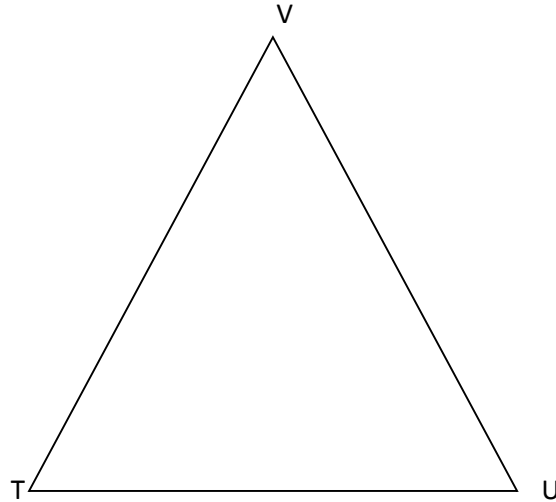
- A. 3.00p.m
- B. 3.00 a.m

- C. 4.00p.m
D. 4.00 p.m
33. According to survey, three antelopes die every five minutes whereas four are born every minute. What is the difference between the number of antelopes that are born and those that die in one day?
- A. 48 96
B. 1440
C. 5760
D. 2448
34. Construct the triangle JKL such that $JK=JL=5.5\text{cm}$ and $KL = 7\text{cm}$. Draw the bisector of the angle KJL to meet line KL at M.
Which one of the following statements is correct from the construction?
- A. Triangle JML is an isosceles triangle
B. Triangle JKL is the Right angled.
C. Line JM is perpendicular to KL
D. Angle JKL $=45^\circ$
35. The hire purchase price of a T.V set is 40% more than the cash price. The cash price is sh. 20 000
Ochieng paid sh. 2100 as a monthly instalment for 10 months.
How much had he paid as deposit?
- A. Sh. 8 000
B. Sh. 13 000
C. Sh. 1 000
D. Sh. 7 000
36. A factory hires 12 workers to complete a piece of work in 5 hours. If two of the workers do not turn up, how will it take to complete the work?
- A. 60h
B. 30h
C. 24h
D. 6h
37. Construct a triangle XYZ such that side $YZ = 6\text{cm}$ angle $YZX = 50^\circ$ AND ANGLE $zxy = 35^\circ$
WHAT IS THE LENGTH of side XZ?
- A. 8.1cm
B. 10.4cm
C. 4.6cm
D. 3.4cm
38. Which one of the following expression is the simplest form of
 $\frac{4(tq + t) + 2(3q - 2tq)}{2(t + q) + 2q}$
- A. $\frac{2t + 3q}{t + 2q}$
B. $\frac{2tq + 4t + 3q}{t + 3q}$

C. $\frac{2tq + t + 6q}{2t + 3q}$

D. $\frac{2t + 3q}{t + 3q}$

39. Construct the circle that sides of triangle TUV below.



What is the radius of the circle?

- A. 7 cm
- B. 6.3cm
- C. 3.3cm
- D. 5.8cm

40. A shopkeeper bought 5 trays of eggs at sh. 120 per tray. Ten eggs broke and she sold the rest at sh. 6 per egg.

If a tray holds 30 eggs, what percentage profit did she make?

- A. 50%
- B. 40%
- C. $28 \frac{4}{7} \%$
- D. 240%

41. Muli spent $\frac{3}{4}$ of his salary on food and $\frac{2}{5}$ on rent. He was left with sh. 2100.

What was his salary?

- A. Sh. 3000
- B. Sh. 3675
- C. Sh. 4900
- D. Sh. 7000

42. Mambo took 30 minutes to cycle from home to the market at an average speed of 5 metres per second. After staying at the market for 20 minutes, he cycled back and took 25 minutes to reach home. What was the average speed, in metres per second for the whole journey?

- A. $5\frac{5}{11}$ metres per second
 B. 4 metres per second
 C. 2 metres per second
 D. 6 metres second
43. One cuboid measures 2m by 4m by 3m. Another cuboid measures 10cm by 8cm by 5cm. What is the difference in their volumes in cubic metres?
 A. 23.6
 B. 23.96
 C. 23.996
 D. 23.9996
44. A shopkeeper bought a shirt for sh. 300. He then sold it for sh. 360 after giving a discount of 10%. What percentage profit would he have made if no discount was given?
 A. $33\frac{1}{3}\%$
 B. $27\frac{7}{9}\%$
 C. 25%
 D. 20%
45. The figure below represents a quarter of a circle centre O. The radius of the circle is 7 cm.
- What is the area of the shaded part?
 (Take $\pi = \frac{22}{7}$)
 A. 24.5cm^2
 B. 14cm^2
 C. 28.5cm^2
 D. 63cm^2
46. Sara bought pawpaws, oranges and mangoes. The number of pawpaws bought was 8 more than the number of oranges. The number of mangoes was twice the total number of oranges and pawpaws. The total number of fruits bought was 48. If the number of oranges was x, which of the equations below can be used to find the number of oranges bought?
 A. $3x + 24 = 48$
 B. $6x + 16 = 48$
 C. $6x + 24 = 48$
 D. $4x + 16 = 48$
47. A shopkeeper bought sodas as follows:
 Two crates of 300ml @sh. 415
 Three crates of 500ml @sh. 563
 Five crates of one litre @sh. 415
 He then spent sh. 50 on transport. If he had five thousand shillings, how much money did he remain with?
 A. Sh. 356
 B. Sh. 406
 C. Sh. 3557

D. Sh. 4644

48. A cyclist rode for two hours at an average speed of 12km/h. He rested for 30 minutes and continued for one hour at an average speed of 8km/h.

Which one of the graphs represents the cyclist's journey?

49.

Which one of the shapes below should be drawn in the blank box to continue with the pattern above?